

BedBoard — PRD & Tech Spec

Sep 23, 2026 · @Someone

Summary

BedBoard is a real-time shelter bed availability board for the people who place clients: Here4You hotline staff and street outreach teams. Shelter staff update open beds in under 10 seconds from a phone; placers see live counts filtered by who a bed can serve (women, families, pets, accessibility) and can place a 60-minute hold.

BedBoard is a tool for the county's existing system, not a public queue. Unhoused people never sign up for it; they reach it through Here4You and outreach workers, which keeps the county's prioritization rules intact.

Owner: Mitansh Mittal (SCU) · **Target MVP:** 6 weekends · **Target pilot:** 3–5 shelters in Santa Clara County, spring 2027 · **License:** Apache 2.0, built on the open-source Finding A Bed Tonight platform · **Honest status:** highest-impact and highest-risk of the three projects, because it needs county buy-in before launch.

Problem and evidence

Santa Clara County already centralizes shelter referrals, and that system works better every year. The remaining gap is speed and visibility: placers still need to know, right now, which bed fits this specific person.

Fact	Number	Source
People experiencing homelessness, Jan 2025	10,711	County
Shelter beds added since 2023	364	PIT summary
Shelter bed utilization	74% in 2023 → 88% in 2025, credited to Here4You improvements	PIT summary
Chronically homeless residents	4,650, up 21% from 2023	Local News Matters

Fact	Number	Source
Here4You hotline staffed hours	9 am–7 pm, 7 days	County shelter page

What works elsewhere: San Diego's Shelter Ready lets outreach workers find and reserve beds in minutes instead of hours or days ([10News](#)). It cost the DA's office \$300,000 to fund. An open-source alternative, [Finding A Bed Tonight](#), estimates placement time falls from 55–140 minutes to 25–45 minutes with live availability and a short bed hold.

Build vs. adopt: Finding A Bed Tonight already has bed search, holds, domestic violence protections and an HMIS bridge to Clarity, the county's HMIS vendor ([SCC HMIS](#)). BedBoard should deploy and localize it for Santa Clara County, not rewrite it. Your original work is the county pilot layer: SMS bed updates for low-tech shelters, the Here4You view, Spanish/Vietnamese, and the adoption itself.

Users and personas

BedBoard has three user groups, and the shelter coordinator is the one who makes or breaks it: if updating takes more than 10 seconds, the data goes stale and placers stop trusting it.

Persona	Role	Job to be done	Must-have
Denise, shelter coordinator	Runs intake at a 60-bed shelter; shared desktop and personal phone	Keep bed counts right with no extra work	Update by SMS ("W 3 M 1") or 3 taps; no new passwords per shift
Luis, Here4You call agent	Answers the hotline 9–7	Match a caller to a fitting open bed while on the phone	One screen, filters (gender, family, pets, ADA), freshness shown on every count
Priya, HomeFirst outreach worker	Works encampments and vehicles, often at night	Find and hold a bed before the client changes their mind	Mobile PWA, works offline, one-tap hold with countdown

Persona	Role	Job to be done	Must-have
County OSH analyst	Oversees shelter system	See unmet demand and utilization trends	Weekly export, no client-level data

People experiencing homelessness are the beneficiaries, not users. They never create accounts, and no client name is stored in BedBoard.

Goals, non-goals, success metrics

Success is fewer minutes from "this person wants a bed" to "this person has a bed," measured on the pilot shelters.

Goals

1. Show live bed counts by population type for every pilot shelter, with freshness on each number.
2. Let placers hold a bed for a set time while transport is arranged.
3. Keep shelter updates under 10 seconds, including by plain SMS.
4. Feed bed inventory into the county's existing HMIS rather than creating a parallel record.

Non-goals

- Deciding who gets a bed. Prioritization stays with the county's coordinated entry rules.
- Public-facing bed counts or a self-service waitlist.
- Client case management or storing client identity.
- Domestic violence shelter locations in any visible form.

Metric	Baseline	Pilot target (90 days)	How measured
Median time from search to confirmed hold	Measure in weeks 1–2 via staff survey	50% reduction	Hold timestamps vs baseline log
Share of counts "fresh" (updated < 2 h) at 8 pm	n/a	≥ 90%	Snapshot age at nightly check
Holds that convert to arrivals	n/a	≥ 75%	Hold status CONFIRMED / total

Metric	Baseline	Pilot target (90 days)	How measured
Open beds left unused overnight at pilot sites	Shelter reports	25% fewer	Nightly empty-bed count
Coordinator update time	n/a	≤ 10 s median	Client-side timing
Staff satisfaction	n/a	≥ 4 / 5	Monthly 3-question survey

Functional requirements

"Base" means the requirement already exists in Finding A Bed Tonight (FABT) and needs configuration and testing; "New" means BedBoard builds it.

ID	Requirement	Acceptance criteria	Source	Priority
BB-1	Bed search by population type and needs	Filters: single men, single women, families, youth 18–24, veterans, pets, ADA; results ranked by fit then distance	Base	P0
BB-2	Freshness on every count	FRESH < 2 h, AGING 2–8 h, STALE > 8 h, shown as label + color + text (not color alone)	Base	P0
BB-3	Soft hold with auto-expiry	HELD → CONFIRMED / CANCELLED / EXPIRED; default 60 min, admin-configurable; count drops immediately	Base	P0
BB-4	Coordinator 3-tap update	Web/PWA update of each population count in ≤ 3 taps	Base	P0
BB-5	SMS bed update	Registered coordinator phone texts "W 3 M 1 F 0" (women/men/family) → parsed, confirmed back, snapshot saved; unknown numbers rejected	New	P0

ID	Requirement	Acceptance criteria	Source	Priority
BB-6	Stale-count nudge	If a site is STALE at 5 pm or 9 pm, text the on-shift coordinator; escalate to site lead after 30 min	New	P0
BB-7	Here4You wallboard	Read-only large-screen view for the call center: all pilot sites, counts, freshness, active holds	New	P1
BB-8	DV opaque referral	DV shelters never appear by name or address; referral via time-limited token + warm handoff call	Base	P0
BB-9	HMIS bed inventory push	Nightly push of bed inventory to Clarity; client data never leaves BedBoard (none stored)	Base, configure	P1
BB-10	Spanish + Vietnamese UI	All coordinator and placer screens translated and reviewed	New	P1
BB-11	Surge / cold-weather mode	Admin activates overflow capacity (e.g. winter warming sites) and all placers see it	Base	P1
BB-12	Unmet demand log	Placer logs a search that found no fitting bed (population type only, no identity)	Base	P1
BB-13	Accessibility	WCAG 2.1 AA; tested with screen reader and 200% zoom	Base, verify	P0

UX and core flows

The core loop is: coordinator updates, placer searches, placer holds, client arrives, coordinator confirms. Every step must work on a cheap Android phone with a weak signal.

```
sequenceDiagram
    participant C as Coordinator
    participant B as BedBoard
    participant P as Placer
    C->>B: SMS "W 3 M 1 F 0"
    B-->>C: "Saved: 3 women, 1 men, 0 family"
    P->>B: Search: single woman, pet
    B-->>P: Site A: 2 fresh (12 min ago)
    P->>B: Hold 1 bed, 60 min
    B-->>C: "Hold: 1 women's bed, arrives by 8:40"
    C->>B: Confirm arrival
    B-->>P: Hold confirmed
```

The coordinator hears about every hold in real time, so they never give that bed to a walk-in.

Screens

Screen	User	Key elements
Search	Placer	Need chips (women, family, pets, ADA), list sorted by fit, freshness badge, "Hold" button
Hold	Placer	Countdown, site address and intake phone, cancel, directions
Update	Coordinator	One row per population type, +/- steppers, big Save; last update time
Holds inbox	Coordinator	Incoming holds with ETA; Confirm / No-show
Wallboard	Here4You	Grid of sites × population types, auto-refresh 30 s, stale sites flagged
Admin	County / lead	Sites, users, hold duration, surge mode, exports

Failure states that must be designed, not left to chance: no fresh beds anywhere (show newest AGING counts + intake phone numbers), hold expires in transit (warn at 10 min left, one-tap extend once), phone offline (queue the update, show "will send when online").

System architecture

BedBoard = the FABT "Lite" tier (Spring Boot backend, React PWA, Postgres only) plus a small BedBoard sidecar service that adds SMS, nudges and the wallboard. Keeping BedBoard code in a sidecar means upstream FABT upgrades stay easy.

```
flowchart LR
    PWA[FABT React PWA] --> API[FABT Spring Boot API]
    API --> DB[(Postgres + RLS)]
    SMS[Coordinator SMS] <--> TW[Twilio]
    TW --> SC[BedBoard sidecar]
    SC -->|REST, service account| API
    SC --> NJ[Nudge scheduler]
    WB[Here4You wallboard] -->|SSE| API
    API -->|nightly| HMIS[Clarity HMIS]
```

The sidecar never touches the database directly; it uses FABT's API with a scoped service account, so all FABT security rules still apply.

Component	Tech	Responsibility
FABT API	Java 25, Spring Boot 4 (upstream)	Search, holds, snapshots, DV protections, HMIS bridge
FABT PWA	React (upstream)	Placer and coordinator UI, offline queue
BedBoard sidecar	Python 3.12, FastAPI	SMS parse + confirm, nudges, wallboard config, translations
Database	Postgres 16 with row-level security	Multi-tenant isolation; DV rows invisible without dvAccess
Real-time	Server-sent events	Wallboard and search update within 5 s of a change
Auth	OAuth2/OIDC (county SSO if available), MFA for admins	Role-based access: coordinator, placer, CoC admin

Why fork-and-extend instead of greenfield: FABT already spent 65+ archived spec changes on hard problems (hold concurrency, DV privacy, HMIS export, accessibility). Resolving them solo would take a year and be less safe. Contributing SMS updates back upstream also gives you visible open-source commits on a real civic project.

Data model

FABT stores availability as append-only snapshots, never edited in place, which gives a full audit trail and trend data for free. BedBoard adds only the tables its sidecar needs.

Upstream (FABT) core entities, simplified

Entity	Key fields	Notes
tenant	id, coc_code (CA-500 for Santa Clara County CoC)	One tenant for the county
shelter	id, tenant_id, name, address, geom, is_dv, pets_ok, ada	is_dv rows protected by RLS; address redacted by policy
bed_capacity	shelter_id, population_type, beds_total	Population types: single adult, family, women only, veteran, youth
availability_snapshot	shelter_id, population_type, beds_available, recorded_at, recorded_by	Append-only; latest row = current count
reservation	id, shelter_id, population_type, status, held_by, expires_at	HELD / CONFIRMED / CANCELLED / EXPIRED
unmet_demand	tenant_id, population_type, needs[], ts	No client identity

BedBoard sidecar tables

```

CREATE TABLE coordinator_phone (
  user_id uuid NOT NULL,          -- FABT user
  shelter_id uuid NOT NULL,
  phone_hash bytea UNIQUE NOT NULL, -- HMAC of E.164 number
  phone_encrypted bytea NOT NULL,  -- needed to send nudges
  shift text,                      -- 'day' | 'evening' | 'overnight'
  active boolean DEFAULT true
);
CREATE TABLE sms_update_log (
  id bigserial PRIMARY KEY, phone_hash bytea, shelter_id uuid,
  raw_parsed jsonb,              -- {"W":3,"M":1,"F":0}, not raw text
  result text,                   -- 'saved' | 'rejected' | 'parse_error'
  ts timestamptz DEFAULT now()
);
CREATE TABLE nudge (
  id bigserial PRIMARY KEY, shelter_id uuid, level smallint, -- 1
  coordinator, 2 lead
  sent_at timestamptz, resolved_at timestamptz
);

```

The CoC code CA-500 should be confirmed with the county before setup.

Integrations and standards (HMIS, HSDS)

BedBoard should plug into the county's systems, not sit beside them. Each integration below needs a named county or agency owner before it goes live.

System	Owner	Integration	Direction	Phase
Clarity HMIS (SCC HMIS)	County OSH / Bitfocus	Nightly bed inventory push (HUD HMIS element 2.07) via FABT's HMIS bridge	BedBoard → HMIS	Pilot, after county approval
Here4You call center	Bill Wilson Center (Here4You)	Wallboard login; hold notifications to shelters	Read + hold	Pilot
Outreach teams (e.g. HomeFirst)	Provider agencies	PWA accounts per worker	Read + hold	Pilot

System	Owner	Integration	Direction	Phase
Twilio	BedBoard	Coordinator SMS in/out	Both	MVP
Open Referral HSDS 3.0	Standard	FABT's HSDS bed-availability extension for data exchange	Export	Later
OpenNow (sister project)	You	Never shares bed counts publicly; only a "call Here4You" pointer	None	n/a

Standards to follow: HUD HMIS Data Standards for any inventory export, HSDS 3.0 for service descriptions, WCAG 2.1 AA for all screens (ADA Title II applies to county-used web tools).

Privacy, security, compliance

BedBoard stores zero client personal information, which removes most of the legal risk. The remaining risks are domestic violence shelter exposure and account misuse.

Area	Requirement
Client data	None stored: holds reference a population type and placer, never a client name, DOB or photo
DV shelters	FABT opaque referral: DV sites hidden from search, address redacted by policy, referral by time-limited token + warm handoff call (designed to support VAWA/FVPSA confidentiality)
Access control	Postgres row-level security per tenant; roles for coordinator, placer, CoC admin; MFA for admins
Accounts	Invite-only; deactivated within 24 h when staff leave (agency admin responsibility, audited monthly)
SMS updates	Accepted only from registered coordinator numbers; replies never contain DV information

Area	Requirement
Audit	Every hold, update and admin action logged, tamper-evident (FABT audit chain)
Hosting	U.S. region; encryption in transit and at rest; daily backups, 30-day retention
Agreements	Data use agreement with the county before any HMIS connection; MOUs with each pilot shelter
Security testing	OWASP ZAP baseline scan before pilot; dependency scanning in CI

Legal note: a student project cannot sign county data agreements alone. A fiscal sponsor or partner nonprofit (e.g. through the Ciocca Center or a civic tech group) should be the contracting party.

Tech stack, infra and cost

The pilot runs for about \$18–\$40 a month, because FABT's Lite tier needs only Postgres and the SMS volume is small. Compare that with San Diego's \$300,000 app.

Layer	Choice
Base platform	Finding A Bed Tonight, Lite tier (Spring Boot + React PWA + Postgres)
Sidecar	Python 3.12 + FastAPI, Docker
Hosting	Oracle Cloud Always Free (FABT documents this deploy path) or a \$20/month VM
SMS	Twilio local or toll-free number
Observability	FABT's Prometheus + Grafana; Sentry for the sidecar
CI/CD	GitHub Actions; Playwright + Karate tests inherited from FABT

Monthly cost, 5-shelter pilot (Twilio July 2026 [pricing](#)):

Item	Assumption	Monthly cost (USD)
Coordinator SMS	5 sites × 3 updates/day × 30 days, 1 in + 1 confirm each, ~\$0.012 per segment	~\$11
Hold + nudge texts	~300 per month	~\$4
Phone number	1	\$1.15–\$2.15
Hosting	Free tier or small VM	\$0–\$20
Domain + email		~\$2
Total		~\$18–\$40

Funding model (no profit): BedBoard is free software. The county or a partner pays only actual hosting and SMS costs, billed at cost with receipts; no license fee, markup or support contract profit. Mitansh's time is pro bono. If the county's rules require paid work (public agencies often cannot accept unpaid volunteer work that matches paid duties), pay is the local minimum wage through the fiscal sponsor. Costs are published quarterly. This is not legal advice; confirm with the fiscal sponsor's counsel before any county agreement.

Testing and QA

The critical invariant is that a bed is never double-held and a DV shelter is never exposed; both get automated tests that block every release.

Layer	What	Target
Inherited FABT suite	Playwright UI, Karate API, RLS and DV canary tests	All green on every sidecar change
SMS parser	"W 3 M 1 F 0", "w3 m1", "women 3", Spanish/Vietnamese variants, garbage input	100% of spec'd formats; unknown input returns help text
Concurrency	20 placers hold the last bed at once	Exactly 1 succeeds

Layer	What	Target
DV leak test	Search as every non-DV role, check API responses and SMS replies	Zero DV names or addresses
Nudge timing	Clock-controlled tests across DST	Fires at 5 pm and 9 pm local, escalates at +30 min
Tabletop pilot	2-hour simulation with Here4You staff and 2 coordinators using fake sites	All flows done unaided; list of friction points
Accessibility	axe-core in CI, manual screen-reader pass	Zero critical violations

Pilot gate: county OSH sign-off, signed shelter MOUs, and a passing tabletop exercise.

Milestones and rollout

Discovery comes before code: if the county already has live bed tracking in Clarity, BedBoard pivots to whatever gap the interviews reveal.

Target date	Milestone	Exit criteria
Oct 2026	Discovery: 8 interviews (Here4You agents, 3 coordinators, 2 outreach workers, OSH staff)	Written list of how beds are tracked today and the top 3 pains
Oct 2026	Contact FABT maintainer	Agreement on contributing the SMS module upstream
Nov 2026	Go / pivot decision	County contact willing to sponsor a pilot, or a documented pivot
Nov–Dec 2026	Build: 6 weekends (FABT deploy, sidecar, SMS, nudges, wallboard, translations)	All PO requirements pass tests
Jan 2027	Tabletop exercise with Here4You + 2 shelters	Pilot gate passed
Feb 2027	Pilot starts at 3–5 shelters	Coordinators updating daily

Target date	Milestone	Exit criteria
May 2027	90-day pilot review with OSH	Metrics reported; expand, change or stop

Stop rule: if fewer than 60% of pilot counts are fresh at 8 pm after 4 weeks, pause and fix the update flow before adding sites.

Partnerships and adoption

Adoption is a relationship problem, not a code problem. The pitch to every partner is the same: less phone tag, no new data about clients, free and open source.

Partner	Why they care	Ask	Path in
County Office of Supportive Housing	Owns the shelter system and Here4You contract	Pilot sponsor; data use agreement	Intro via Destination: Home or SCU Unhoused Initiative faculty
Bill Wilson Center (runs Here4You)	Agents spend time calling shelters	Discovery interviews; wallboard trial	Their Here4You program lead
Pilot shelters (3–5)	Fewer calls, fewer empty beds overnight	Daily updates for 90 days	Through OSH and the Unhoused Initiative
HomeFirst outreach	Night placements are hardest	Outreach worker testers	Through OSH
Finding A Bed Tonight maintainer	Wants real deployments	Code review, upstream SMS module	GitHub
SCU Ciocca Center / Miller Center	Social venture support	Fiscal sponsorship, mentorship	Bronco Ventures networking nights

Your story: "Led discovery with county shelter staff and deployed an open-source bed availability system across X shelters, cutting placement time by Y%." Even the discovery alone is a strong story if the pilot never launches.

Risks and open questions

The top risk is that the county already has what BedBoard offers, which discovery answers in October at almost no cost.

Risk	Likelihood	Impact	Mitigation
County already tracks live beds in Clarity	Medium	High	Discovery first; pivot to the gap found (e.g. SMS updates feeding Clarity)
No county sponsor	Medium	High	Start with one agency's own shelters as a private pilot
Coordinators stop updating	High	High	SMS updates, nudges, share wins ("your beds filled 2 h faster")
DV location leak	Low	Critical	Inherited FABT protections + release-blocking leak tests
Upstream FABT changes break sidecar	Medium	Medium	Pin versions; sidecar talks only to the public API
Legal contracting as a student	High	Medium	Fiscal sponsor as contracting party

Open questions

- ☐ Does Here4You already see live bed counts, and how often are they updated?
- ☐ Which 3–5 shelters have the most walk-in churn and would benefit first?
- ☐ Would OSH prefer BedBoard to push into Clarity, or to read from it?
- ☐ Who at SCU can make a warm introduction to OSH?

Sources

- [County 2025 PIT count release](#)
- [2025 PIT count executive summary](#)
- [County emergency shelters page](#)
- [Bill Wilson Center, Here4You](#)
- [Santa Clara County HMIS, Here4You resources](#)
- [Finding A Bed Tonight \(GitHub\)](#)

- [10News on Shelter Ready](#)
- [Local News Matters on chronic homelessness](#)
- [Twilio US SMS pricing](#)